These were the reasons why he argued that it was not economic growth itself that created environmental problems but how it was achieved. The new production technologies had a far greater environmental impact than the ones they replaced. As *an example*, Commoner looked at farm technologies. He pointed out that the traditional *fertilising system* of farms, where animals provided the manure for fertilising the land, had been interfered with by the use of feedlots, where animals were confined in small areas whilst being fattened up for market rather than leaving them grazing on pastures. The resulting heavy concentration of manure placed undue strain on a small area of land which could not naturally deal with so much waste. The waste, therefore, tended to pollute underground and nearby waterways. Animals in feedlots were fed on grain, and the land used to grow the grain was depleted of nutrients; so that farmers had to resort to artificial fertilisers, especially nitrogen, that created their own pollution problems because some of the chemicals used ended up in the waterways.

Similarly, the use of *pesticides* also enabled farmers to get higher yields from smaller land areas, but at an environmental cost. Pesticides such as DDT also polluted waterways, and killed or harmed other insects and animals (and sometimes humans) that were not originally targeted. While artificial fertilisers depleted the soil of naturally occurring nitrogen-fixing bacteria, pesticides killed off the pests' natural predators, and the pests themselves built up resistance to the pesticides. This ensured continuing dependence on the new chemicals and the need for ever-increasing amounts to be used.

In a*nother example* Commoner talked about the replacement of soaps by detergents. He estimated that the production of the active agent of detergents required three times as much energy as soap. The burning of fuel and high- temperature reactions needed during manufacture of detergents added to air pollution. Not only did the manufacture of detergent subject the environment to greater stress than soap, but its disposal created a whole new set of problems. The original detergents did not biodegrade in the environment, and they created mountains of foam in waterways. The new generation of detergents developed to solve this problem did not produce foam but were more toxic to the fish in the waterways. Also, the phosphate in the detergents stimulated algal growth which could choke rivers or stress